

Researched-based Strategies for Designing a Great Research Poster

ME495



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COLLEGE OF ENGINEERING
PROGRAM IN TECHNICAL COMMUNICATION
UNIVERSITY OF MICHIGAN

Today's Learning Objectives:

1. Apply organizational features to create an effective poster
 - a. Create logical headers and supporting text/visuals
 - b. Develop argumentative structures
2. Demonstrate effective use of language, contextualized within the genre
 - a. Maintain professional and appropriate tone
 - b. Use content-specific persuasive style
3. Illustrate data effectively
 - a. Create large, clear, readable visuals that support arguments
 - b. Link visual evidence to tell a complete [data story](#)

Suggestion:
create a group google slide
to take notes, create placeholders,
assign tasks, etc.

**You will also create a 3-page handout;
create that Google Doc now**

Friday, 9/18 is poster workshop day; bring a draft!

Trade posters with another student

Receive feedback from your peers

Receive feedback from the instructional team

Ask questions

Finish your poster under guided work time

Tentative poster presentation schedule

There is still some movement in the labs so this may change!

	sec	day	lab start	lab end	pres start	pres end
9/22	3	Tu	11:30 AM	2:20 PM	11:30 AM	1:30 PM
	4	Tu	2:30 PM	5:20 PM	3:20 PM	5:20 PM
	5	Tu	5:30 PM	8:20 PM	5:30 PM	7:30 PM
9/23	6	We	11:30 AM	2:20 PM	11:30 AM	1:30 PM
	7	We	2:30 PM	5:20 PM	3:20 PM	5:20 PM
	8	We	5:30 PM	8:20 PM	5:30 PM	7:00 PM
9/24	9	Th	2:30 PM	5:20 PM	3:20 PM	5:20 PM
	10	Th	5:30 PM	8:20 PM	5:30 PM	7:30 PM

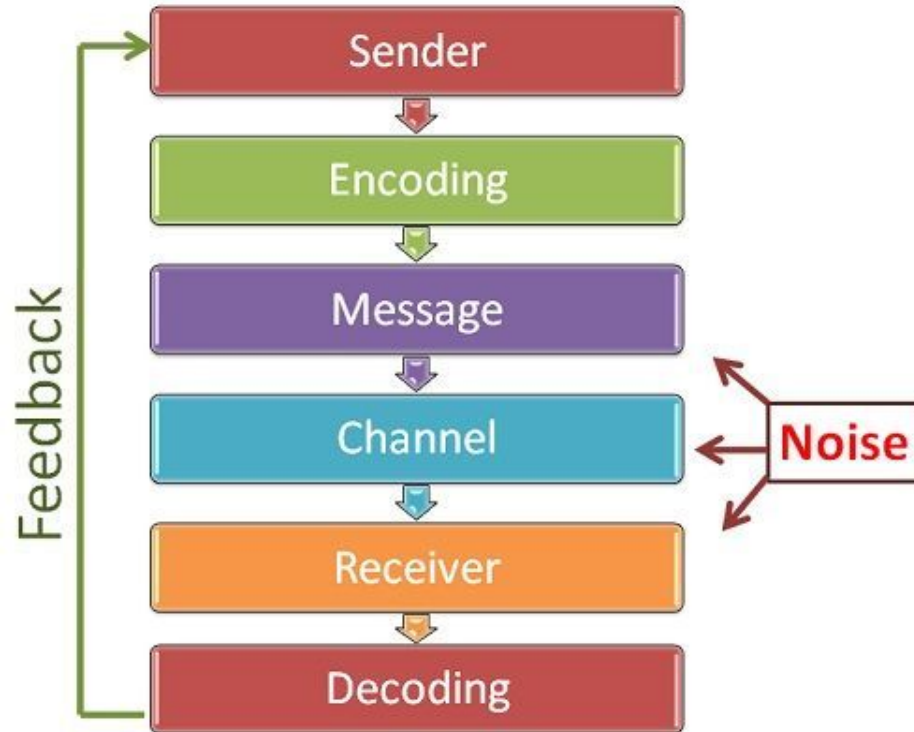
You have 10 minutes to present to experts and non-experts, followed by Q&A

Bring 4 printed copies of the poster & handout

**Some basic principles
will help you succeed in ME495
and beyond**

- Understand the communication process
- Follow Doumont's rules
- Consider perceived cognitive load

A lot needs to happen for effective communication to occur



Doumont's 3 rules of effective communication are universal

1. adapt to your audience



2. maximize the signal-to-noise ratio



3. use effective redundancy





Mental effort = disengagement

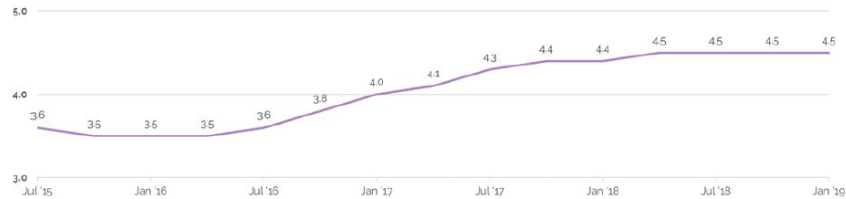
1. Guide your viewer's eye by overlaying visual annotations
2. Do the work for your audience

Image credit: Mike Morrison, the #betterposter movement. In short, meet people where they are. Don't confuse them.

Do the work for your audience

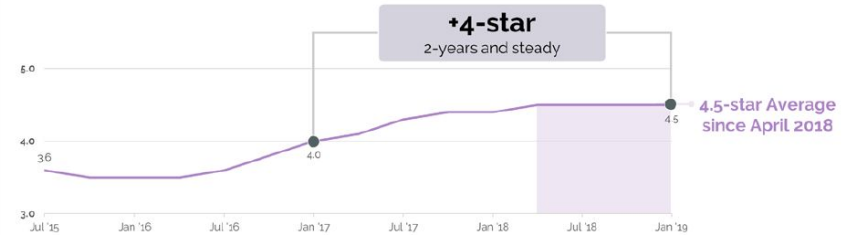
XYZ quality scores: steady rise and stable average

XYZ QPC National Average Star Rating
July 2015 – January 2019



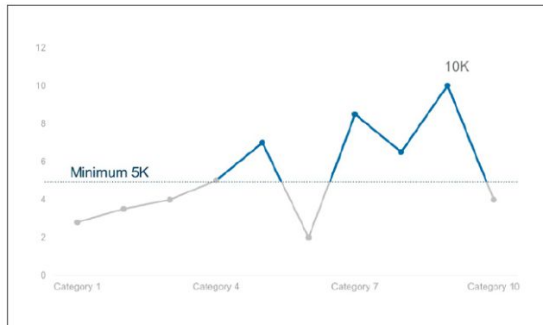
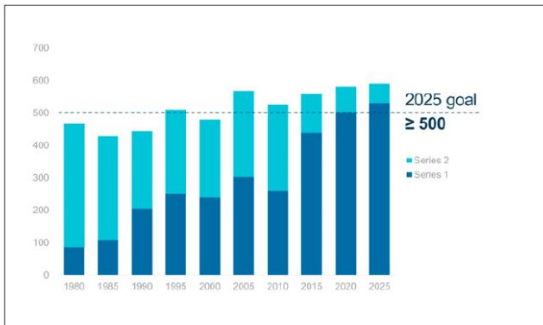
XYZ quality scores: steady rise and stable average

XYZ QPC National Average Star Rating
July 2015 – January 2019



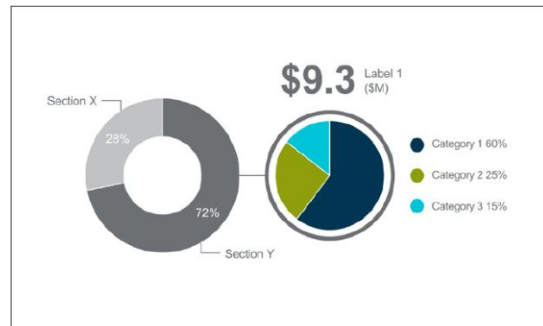
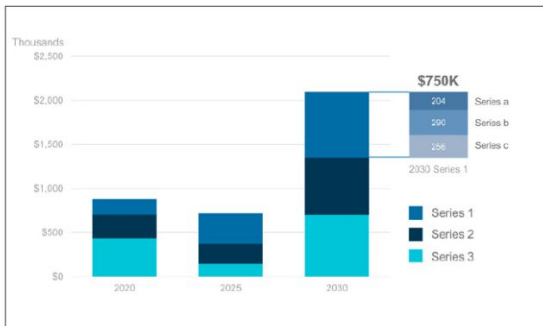
DELINEATING DATA SHOWS SHORTFALLS AND SURPLUSES

Adding a simple horizontal line to indicate a baseline turns a simple chart into a clear visualization of where benchmarks are being met and where they are not.



DISSECT DATA BY EXPLODING IT OUT

To provide additional information on one data set, create a secondary element on the chart where you highlight subcategories with colors and graphics.



Perceived cognitive load stands between your message & your audience



Patterned Inflatable Silicone Membrane for Vaneless Vents

MEUS WINTER 2019

GM

Introduction

Conventional automotive vents have not yet been able to adopt recent trends in stream line aesthetics and minimalist design due to need for manual mechanism of flow control. Recently automatically control vents have been developed; however, the its inner mechanism is very intricate. This poster presents a new approach to directing airflow by using vaneless vents. The current vaneless vent has the capability to redirect air, but only vertically. The patterned inflatable membrane is integrated on the current vaneless vent to redirect air horizontally when inflated.



Overview

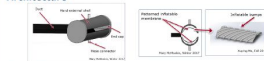
Goal: To develop and characterize the performance of a patterned inflatable silicone membrane able to produce bumps capable of directing air laterally to be integrated into the current vaneless vents.

Objectives:

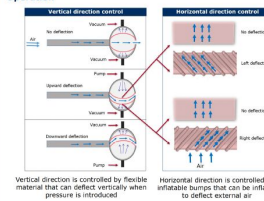
- Experimentally characterize materials and select a final material to produce vaneless vents that can redirect airflow horizontally with inflatable features to a desirable degree
- Analyze and characterize inflation height of final membrane and its relation to membrane geometry
- Validate the shape and dimensions of inflatable membrane needed to redirect transversal airflow

Vaneless Vents

Architecture



Operation



Masked Drop Casting Process



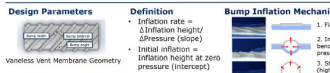
Mask and Material Studies

Inflatable Membrane

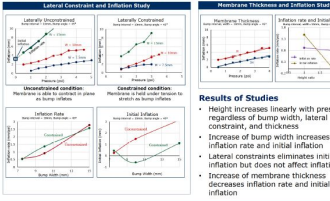
Inflatable Membrane Evaluation Metrics

Material	Thickness	Inflation	Performance Evaluation	Cost	Manufacturability
Polysiloxane	0.5mm	✓	✓	✓	✓
Polysiloxane	1.0mm	✓	✓	✓	✓
Polysiloxane	1.5mm	✓	✓	✓	✓
Polysiloxane	2.0mm	✓	✓	✓	✓
Polysiloxane	2.5mm	✓	✓	✓	✓
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Polysiloxane	5.0mm	✓	✓	✓	✓
Polysiloxane	5.5mm	✓	✓	✓	✓
Polysiloxane	6.0mm	✓	✓	✓	✓
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Polysiloxane	7.0mm	✓	✓	✓	✓
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Polysiloxane	8.0mm	✓	✓	✓	✓
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Inflation Characterization



Membrane Inflation Property Study



Experimental Characterization

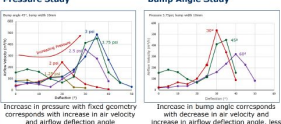


Experimental setup allows constant external airflow, while measuring the air velocity using anemometer. Experimental setup top view: air velocity is measured in terms of deflection angle if set using an angle gauge.

Procedure

- Inflate inflatable membrane inside prototype
- Generate constant air source to go straight through the damper, prototype, and anemometer sensor.
- Record the velocity of air with different angle gauge position

Pressure Study



Conclusion

- Silicone membrane for vaneless vent enables minimalist and aesthetically pleasing ventilation design to be incorporated in future automobiles
- Silicone membrane created using Elastoll 94117 for vaneless vent can be inflated to redirect airflow horizontally to a desirable degree
- Continuous controllable lateral airflow redirection up to 30° can be achieved with fixed geometry by increasing pressure which validates the design
- Lateral constraint does not affect membrane's inflation rate but eliminates initial inflation
- Increase in bump angle will increase airflow redirection angle, but decrease air velocity and less effective for larger bump angle
- Continuously controllable lateral airflow redirection up to 30° can be achieved with fixed geometry by increasing pressure which validates the design

Acknowledgements

Special Thanks

Michigan Automated Scale System (MASS)



SECTION INSTRUCTOR SPONSORS

PROBLEM

The UM/GM Advanced Battery Coalition for Driveables (ABCOD), headed by Professor Wey Lu at the University of Michigan, researches battery technologies for improving electric vehicles. In the development of a novel battery separator technology, they must experiment with many different formulations of chemical solutions, requiring the accurate measurement of the masses of different chemicals. This has proven to be a bottleneck in the research process. Thus, we have been tasked with designing, building, and testing a small and cost-effective machine to automatically measure and dispense batches of different powdered chemicals.

DESIGN PROCESS, CONTEXTUAL FACTORS, AND SOCIETAL IMPACT

Our design reduces the time it takes to measure out chemicals for testing, contributing to the research and development of electric vehicle (EV) batteries. More efficient EV batteries mean lower carbon emissions, acceleration in EV market growth, and less environmental damage. A consequence of our design is that increased EV efficiency would negatively impact the gasoline-powered vehicle market and oil companies.

REQUIREMENTS & SPECIFICATIONS

Priority	Requirements	Specifications	Why
High	Must create batches of powder chemicals with specific mass (accuracy)	Dispensed chemical with ±0.01g accuracy	±0.01g accuracy required for battery separator chemical formulation research
High	Must be automatic	After startup, 0 touches required to measure mass	Aim to reduce human time taken to conduct experiments
High	Must be fast	<15s for 10g <90s for 100g	Quick operation required to save time



MSEC

MSEC NAMRC Conference, Hosted by Purdue University
June 27th, 2022 – July 1st, 2022

Company, Univ and lab logos

Add NSF or other funding agency logo (to the right) →

TITLE OF POSTER ARIAL 88 PT CENTERED ON POSTER SMALL CAPS

Text Box 1 16 inches wide ½ inch from left edge

HEADERS IN BOLD ARIAL 32 PT FONT SMALL CAPS

possible header sections (but author will select based on research)

INTRODUCTION (OR BACKGROUND)
RESULTS
DISCUSSION
CONCLUSIONS
REFERENCES
ACKNOWLEDGMENTS

Body text in Arial 28 point font, upper and lower

All text fully justified

Figures, tables and other graphics should be inserted within the confines of the text box areas shown

Author box centered on poster Author Arial 44 pt bold centered

Affiliations Arial 40 pt italics centered
Upper and lower case
email address

Abstract box centered on poster

ABSTRACT: IN BOLD ARIAL 32 PT FONT SMALL CAPS

200 words max

Body text in Arial 28 point font

All text fully justified

Adjust the box size to fit around the text

Text box 2 centered on poster

HEADERS IN BOLD ARIAL 32 PT FONT SMALL CAPS

Body text in Arial 28 point font, upper and lower

All text fully justified

Figures, tables and other graphics should be inserted within the confines of the text box areas shown

Text box 3 16 inches wide ¼ inch from right edge

HEADERS IN BOLD ARIAL 32 PT FONT SMALL CAPS

Body text in Arial 28 point font, upper and lower

All text fully justified

Figures, tables and other graphics should be inserted within the confines of the text box areas shown

POSTER SUBMISSION FOR REVIEW:

1. Save as a PDF
2. Make sure that the PDF file size does not exceed 15 MB
3. Submit the posters through the ASME conference webtool (and not via email to organizers or ASME staff)
4. E-mail Prof. Binli Starly, NC State (bstarly@ncsu.edu) or Prof. Yong Chen, University of Southern California (yongchen@usc.edu) with any questions about the poster layout or content.
5. Email toolboxhelp@asme.org with any questions about the submissions process.
6. Posters must be submitted by March 1st, 2022
7. Authors will be notified by March 15, 2022
8. You may submit a poster for review even if you did not submit a poster abstract in November 2021.
9. The Copyright Transfer Form must be completed by March 26, 2022
10. If poster is accepted, presenter should be registered by April 15, 2022, to prevent the poster (and abstract) from being withdrawn from the conference.

ACKNOWLEDGE source(s) of funding

Do not forget to include the grant number(s) if the work was funded by a national funding agency.

Remember: audience and purpose override any other rule

Mesa would like to review several possible solutions to the vibration problem and select, based on a broad range of criteria beyond the actual drive shaft design, the best design for the final powertrain. Therefore, you are charged with identifying the four most effective solutions to achieve a 30% reduction in the vibrations.

This project has an extremely short timeline and must have your highest priority. You will be presenting the results of your experimental work and analysis in a poster session in the week of September 21. Because of the significance of this vibration problem, the poster session will be attended by Mesa engineers and executives with diverse technical backgrounds. Please ensure that your poster uses clear visual elements to provide a brief, broadly comprehensible introduction and conclusions alongside the central technical data. In addition, a short, three-page handout summarizing the key findings from your work is due to us in the same week during your poster presentation.

How would you adapt the poster for these scenarios?

1. Poster displayed in a lab during a sponsor visit (no one presents it)
2. Poster emailed to an executive as a summary of work completed
3. Poster presented at an ASME conference
4. Poster presented at a University-wide, interdisciplinary graduate student symposium
5. Poster presented to engineers and marketing experts in a workplace situation

Product design poster: consider the audience

INNOBREW

COFFEE MAKING SYSTEM FOR COMMERCIAL ENVIRONMENTS

DESCI501

Prof. Dhanu Bhai, Prof. Nikos Chronis, Dr. Jonathan Lurtz, Kate Olsen, Audrey Sedal

TEAM DESIGN KITCHEN

Tarun Kumar, Matthew Neilson, Dhairyaash Kulkarni, Korry Bant

CUSTOMER EXPLORATION

SETTING

Commercial environments selling coffee (not as core business)

Business Owner

profit maximization
satisfaction of customer

Attendant

reduced workload
minimized cognitive load
happy customers

Coffee Consumer

freshly brewed coffee
minimum waiting time

vs

vs

CONFLICTS

Cost of Service, Change intervals, Wait times & Freshness

MARKET & POSITIONING

standing time of coffee

Currys high capacity

Innobrew

Bunn medium capacity

Bunn low capacity

frequency of human intervention

Top Down

Bottom Up

- 24.6 million coffee machine sales (2010)

- 18.5 million auto-drip coffee maker

40k

700k+

Assumptions

- 2 machines per business

- Replacement per 6 years

- Target %15 market

- 55,000 restaurants

- 100,000 gas stations

- 624,000 coffee&snack shops

DESIGN SOLUTION

Creating an automated system that ensures availability of fresh coffee while minimizing frequency of human intervention and wasted coffee.

PRODUCT FUNCTIONALITIES

1-Preparing filter holders

2-Loading filter holders

3-Feeding mechanism

4-Brewing position

5-Dumping filter holders

6-Drainage system

prepared by attendant

automatically by innobrew

PRODUCT DETAILS

Parts List

01 - Coffee filter holders

02 - Spring mechanism

03 - Touch screen interface

04 - Rotary arm

05 - Stepper motor

06 - Milk pot

07 - Outer shell

08 - Filter holder rack

09 - Linear actuator

10 - Carriage assembly

11 - Base of rack assembly

12 - Coffee maker body

13 - Dumping structure

14 - Coffee carafe

DESIGN METHODOLOGIES

Operations Modelling

Key Drivers

10 VOLUME filter containers

12 cup brew size

Parameter Analysis

Key Drivers

Nema 17 3.4 kg-cm stepper motor

L16 140mm linear actuator

Stress & Thermal Analysis

Key Driver

Nylon 10 material for rotating arm

Kansei Analysis

Key Drivers

Orange rack cover

Matte body finish

10 container rack size

level	Characteristics		
	color	surface finishing	rack size
-1	orange	Matte	10
1	black	Polished	16

Conjoint Analysis

Key Drivers

Cost of wasted coffee

Freshness of coffee

Innobrew price

BUSINESS OPPORTUNITY

Combined Cost of Attendant + Coffee

- 400 cup daily demand

- 15 sec walking distance (main area to coffee area)

- 1m in filter change time

- \$12/hr attendant wage

- \$0.32 cost per cup

Cost & Running Cash Balance

Material Costs

\$243,70

Manufacturing Costs

\$6,33

Non-Manufacturing Costs

\$37,50

Profit (%6)

\$15,00

Total Sell Price

\$302,54

Outline your poster before you begin creating it

1. Re-read the task letter
2. Consider 'backward design' to sketch out your presentation
 - a. Tell your **data story** (beginning, middle, end)
 - b. Curate a set of visual evidence
3. Plan your oral delivery WITH your visual aid

Your Poster Should Convey the Following:

1. What technical problem you are addressing
2. What approaches you used (experimental, modeling, analysis, design)
3. What key findings you discovered
4. What impact your solution provides

**Let's look at poster elements
top to bottom**

Which title is most effective in telling the audience about the project?

1. IOE 591 Progress Report on Cycle Time
2. Optimizing Ford's Manufacturing Process: IOE 591
3. Six Sigma Application to Ford Manufacturing: Improving Cycle Time
4. Progress Update on Optimizing Cycle Time
5. Ford Progress Update: Six Sigma Optimization has Improved Cycle Time

Titles are important: draw in audiences, searchable, tell a story, signal your argument

1. Topic
 2. Method
 3. Result
- Optional: Client/Sponsor name

Examples:

Optimizing Warehouse Picking Routes Using Simulation to Reduce Fulfillment Time

Improved EV Battery Thermal Management Through Cooling-Channel Optimization and Prototype Testing

Tell us the recommendation at the beginning,
refer to it throughout,
repeat it at the end:

Do this orally, visually, and/or textually.

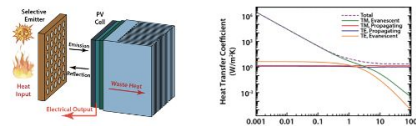
Harnessing Nanoscale Dielectric Coatings to Enhance Near-field Radiative Heat Transfer for Thermo-photovoltaic Applications

Yashar Ganjeh^{1*}, Bai Song^{1*}, Seid Sadat^{1*}, Dakotah Thompson¹, Anthony Florino¹, Victor Fernandez-Hurtado², Johannes Feist², Francisco J. Garcia-Vidal², Juan Carlos Cuevas²

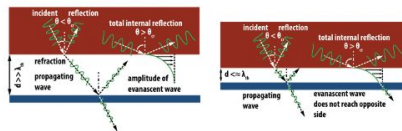
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Motivation

- Thermal management of micro devices
- Thermal microscopy
- Energy conversion, e.g. nano-scale gap thermo-photovoltaics

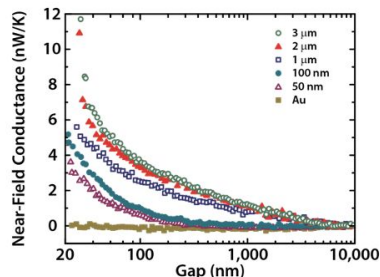


❖ How much can a thin dielectric coating contribute to radiative heat transfer?

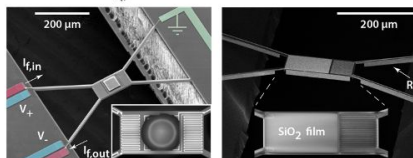
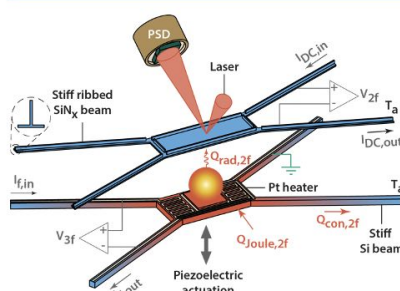


Intuition based on far-field behavior

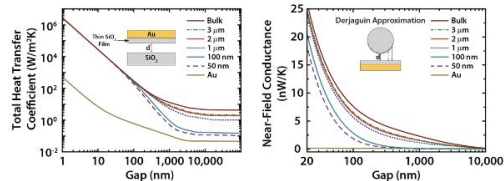
Nanoscale Dielectric Coatings Enhance Near-field Radiative Transport



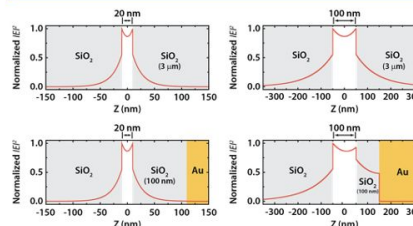
Experimental Setup



Fluctuational Electrodynamics Theory Predicts the Experimental Results



Penetration Depth Scales with Gap-Size



- For small gaps:
 - Mode shapes are identical for different film thicknesses
 - Penetration depth is the same order as the gap-size
 - The modes decay rapidly into the film
- For larger gaps (greater than film thickness):
 - Penetration depth becomes larger
 - The modes decay slowly into the dielectric coating
 - There is a sudden drop in intensity at Au-SiO₂ interface resulting in a diminished heat transfer at the interface

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Don't create a portrait poster for this assignment; remember it will be projected on a (landscape) screen



University of Michigan, Ann Arbor, MI USA



Stochastic Co-encapsulation



Challenge

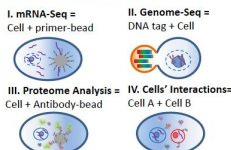


Figure 3: Examples of various droplets-based Single-cell assay.

$$D_{k, \text{Poisson}} = \frac{\lambda^k \exp(-\lambda)}{k!}$$

$$D(k_A \cap k_B, \text{Poisson}) = D_{k_A}$$

where λ is the average number of events.

81.9% > 17.2% > 0.82%

- ✗ Rare cells study
- ✗ Assays required ≥ 3 objects

Figure 4: The conventional cells/beads assays suffer from stochastic pairing process of cells & beads

Result

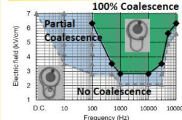
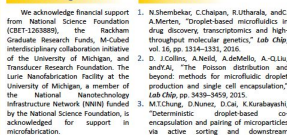


Figure 6: Operation principle of our droplet pairing & merging device. (Side view)

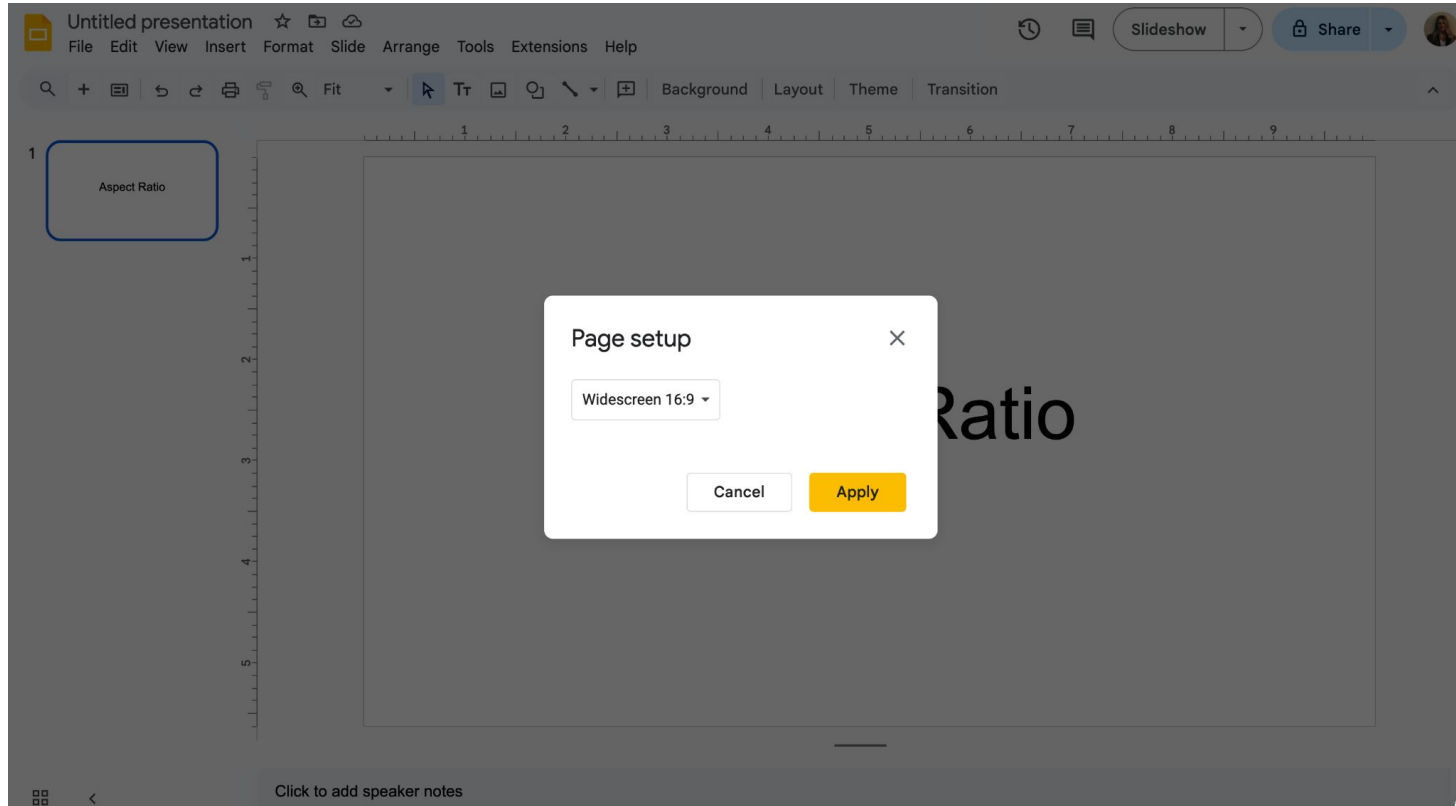
- Buoyance-based Capturing
- Size-based 1-1 Pairing
- Scalable & No back-Pressure
- Electrical Coalescence
- Multi-round merging

Acknowledgement

References



4:3 mimics field of vision, standard television or computer
16:9 HDTV format, most widely used



Test your format and colors on a screen

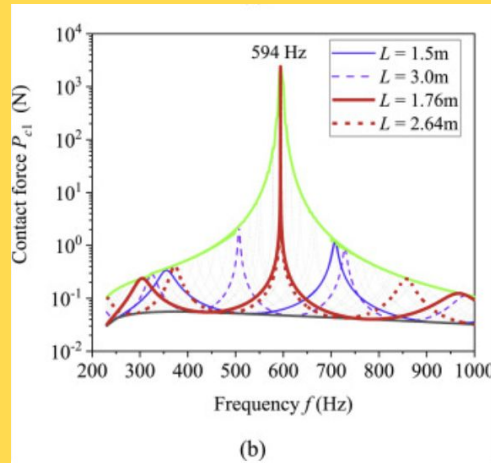


Fig 1: This slide is awful; avoid these things

Which colors are easy to see?

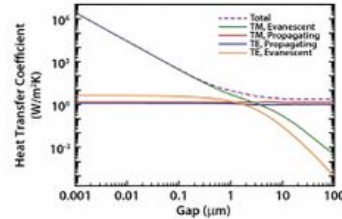
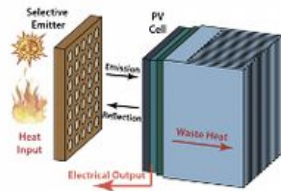
Which have enough contrast?

Is the text large enough to read?

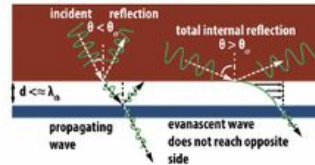
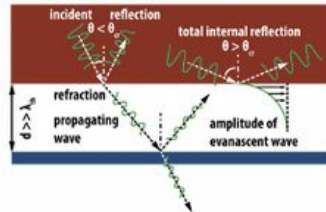
Provide context and the research question aka motivation, objective, goal, introduction

Motivation

- Thermal management of micro devices
- Thermal microscopy
- Energy conversion, e.g. nano-scale gap thermo-photovoltaics



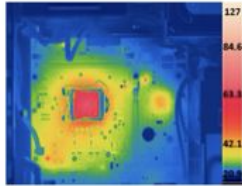
❖ How much can a thin dielectric coating contribute to radiative heat transfer?



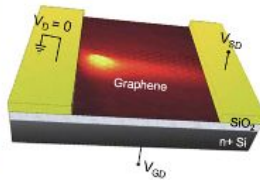
Intuition based on far-field behavior

Introduction

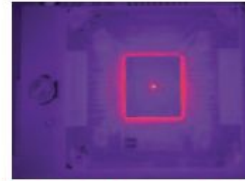
- Continued miniaturization of electronic devices poses serious challenges for nanoscale heat management.
- Classical heat transfer laws break down at the nanoscale.
- Understanding energy transport and conversion is crucial for realizing novel technologies in the near future.



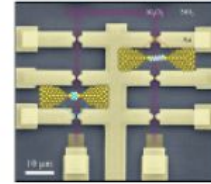
Hotspot of CPU



Hotspot in graphene



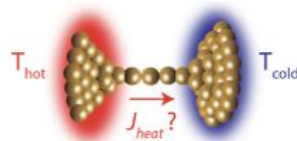
On-chip thermoelectric (Peltier) cooling



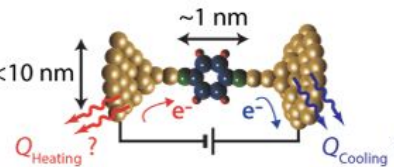
Molecule transistor

Objectives

- What is the fundamental (quantum) limit of heat transfer at the atomic-/nano-scale?
- Is it possible to realize molecular-scale thermoelectric cooling?



Single-atom contact



Molecular junction

Let's look at those examples in detail

- Content-specific headers
- Highly visual
- Intentional use of design and color

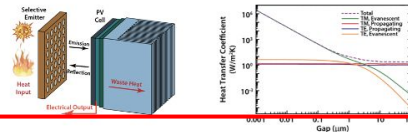
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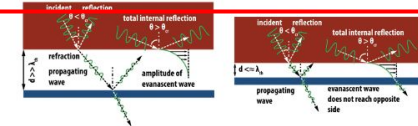
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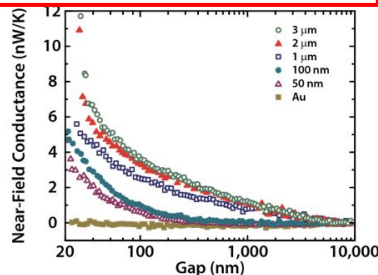


❖ How much can a thin dielectric coating contribute to radiative heat transfer?

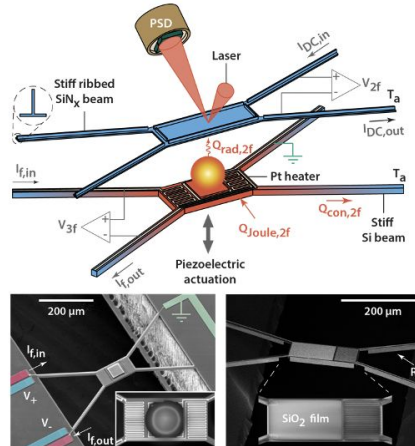


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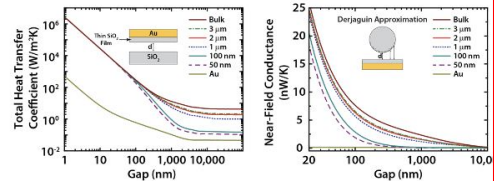
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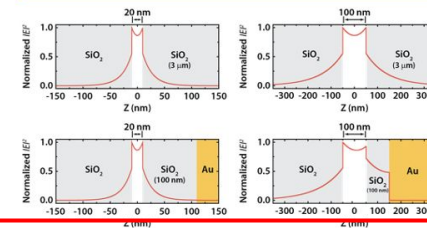
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¹Department of Mechanical Engineering, ²Department of Cell & Developmental Biology, ³Department of Biophysics, ⁴Department of Electrical Engineering and Computer Science, University of Michigan, Ann Arbor, MI USA

Single-cell study reveals cellular heterogeneity. **The key to successful sing-cell research is accuracy and throughput.** Although droplet-based microfluidics has a highest throughput among all the other tools, its efficiency significantly suffers from stochastic encapsulation process which prevents us from many applications. We presents a microfluidic platform for deterministic **co-encapsulation of two and three different particles at a yield as high as 87.0% and 80.1%,** respectively, by integration of active droplet sorting and downstream merging processes. This eliminates stochastic pairing processes for which severely limits the yield to <10%.

This Work

Figure 1: Concept of our droplet manipulation process for cells/beads based assays

The diagram illustrates four single-cell analysis methods:

- I. mRNA-Seq = Cell + primer-bead**: Shows a cell being lysed and its mRNA being captured by a bead with a primer.
- II. Genome-Seq = DNA tag + Cell**: Shows a cell being lysed and its DNA being captured by a bead with a DNA tag.
- III. Proteome Analysis = Cell + Antibody-bead**: Shows a cell being lysed and its proteins being captured by a bead with an antibody.
- IV. Cells' Interactions = Cell A + Cell B**: Shows two cells interacting, with one cell being lysed and its components captured by a bead.

Figure 2: Single-cell analysis reveals cellular heterogeneity.

Figure 3: Examples of various droplets-based Single-cell assay

I. Co-flowing

$D_{\text{Co-Flowing}} = \frac{\lambda^2 \exp(-\lambda)}{2}$

$D(\lambda | n \text{ k-frames}) = D_{\text{Co-Flowing}} \times D_{\text{Passion}}$

where λ is the average number of cells per droplet.

Case $\lambda=0.1$

II. Additive Merging

$D_{\text{Additive Merging}} = \frac{\lambda^2 \exp(-\lambda)}{2}$

81.9% > 17.2% > 0.82%

✗ Rare cells study
✗ Assays required ≥ 3 objects

Figure 4: The conventional cells/beads assays suffer from stochastic pairing process of cells & beads

1. Droplet Generating 2. Sorting 3. Pairing & Merging

Figure 5: 3-D schematics of our microfluidic system composed of droplet generating/sorting and droplet merging devices. Scale bar:100 μ m.

- Buoyancy-based Capturing
- Size-based 1-1 Pairing
- Scalable & No back-Pressure
- Electrical Coalescence
- Multi-round merging

In addition to improve accuracy and efficiency of existing assays...

Stem cells: 0.01-10% in tissues
CTC cells: 0.0001% in blood

A diagram of a cell with various organelles labeled with letters A through S. The cell is an oval shape with a blue border. Inside, there is a large red nucleus with a blue nucleolus. Other organelles include a blue mitochondrion, a green Golgi apparatus, a blue lysosome, a blue vacuole, and a blue vacuole. The labels A through S are placed around the cell, with some pointing to specific organelles. The letter 'S' is located at the bottom left of the cell.

Figure 7: Observed coalescence response under different electric field strength and frequency. The green shaded area indicates all the droplets pairs are merged. Only a fraction of droplets is merged in the blue shaded area. No merging event occurred in white area. The electric field is calculated by the applied voltage V across two electrodes with a separation distance of 15mm.

DC 100 1000 10000 100000
 0 1 2 3 4 5 6 7
 Electric field (kV/cm)
 10 100 1000 10000
 Frequency (Hz)
 100% Coalescence
 No Coalescence

Figure 8: The capability of merging multiple droplets and retrieving them for down-stream analysis

2-Bead Co-encapsulation

Method	Desired Combination (%)
Poisson ($\lambda=0.1$)	~10
Poisson ($\lambda=1$)	~10
This work ($\lambda=0.05$)	87%

3-Bead Co-encapsulation

Method	Desired Combination (%)
Poisson ($\lambda=0.1$)	~10
Poisson ($\lambda=1$)	~10
This work ($\lambda=0.05$)	80.1%

Percentage of Droplets (%)

Legend: ■ Other Combinations, ■ Desired Combination

Figure 9: The histogram shows the success rate of droplets that contain the correct numbers of blue, red, and green fluorescent beads. The data was collected from a total 1176 host droplets in a merge device with 3 repeats.

References

We acknowledge financial support from the National Science Foundation (CBET-1263889), the Rackham Graduate Research Funds, M-Cube interdisciplinary collaboration initiative of the University of Michigan, and Transducer Research Foundation. The Lurie Nanofabrication Facility at the University of Michigan, a member of the National Nanotechnology Infrastructure Network (NNIN) funded by the National Science Foundation, is acknowledged for support in microfabrication.

1. N.Shembenkar, C.Chaijan, R.Utharala, and A.Merten, "Droplet-based microfluidics for drug discovery, transcriptomics and high-throughput molecular genomics," *Lab Chip* vol. 16, pp. 1334-1331, 2016.
2. D. J.Collins, A.Nelid, A.deMello, A.-Q.Liu, and A.L., "The Poisson distribution and beyond: methods for microfluidic droplet production and single cell encapsulation," *Lab Chip*, pp. 3439-3459, 2015.
3. M.T.Chung, D.Nunez, D.Cai, K.Kurabayashi, "Deterministic droplet-based co-encapsulation and pairing of microparticles via active sorting and downstream merging," *Lab Chip*, 2017,17, 3664-3671.

Use parallel structure to eliminate bumps in reading

We aim to improve the passenger experience.



Improve passenger experience

Our goal is to identify the cause of the vibrations.



Identify cause of vibrations

In the end, we propose 4 design modifications.

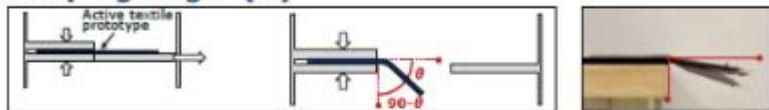


Propose 4 design modifications

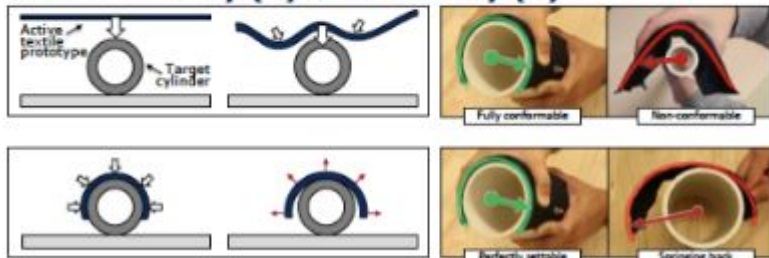
Methods establish ethos

Test Setup and Procedure

Draping Angle (D)



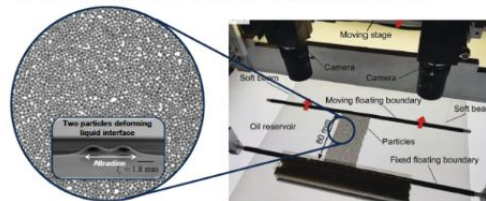
Conformability (C) & Settability (S)



Experimental setup and procedure

System setup:

- Monolayer of polydisperse Styrofoam particles floating at an air-oil interface. Long-ranged capillary attractions between particles exist³, with a characteristic interaction range of the capillary length $l_c = \sqrt{\gamma/(\rho g)}$, with γ being the surface tension, ρ the density of the liquid, and g the gravitational acceleration.
- Displacement applied with floating boundaries with a soft cantilever setup to measure the applied forces. Particle centers and radii were tracked³.

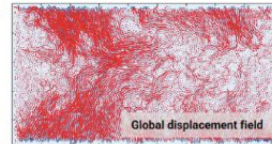


Quasi-static testing procedure:

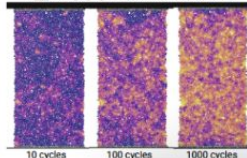
- Mechanical aging:** small amplitude oscillatory deformation was applied for 1000 cycles with 0.3% strain amplitude
- Uniaxial deformation:** tension or compression is applied to the system until failure, with a strain rate on the order of 10^{-5} that minimize viscous drag³.

Dynamics during mechanical aging

- Displacement field:** in stroboscopic images, particles have uneven convection-like current during the oscillatory deformation
- Particle trajectories are irreversible** in all oscillatory cycles, resulting in long-term non-affine displacement



D_{min}^2 during aging, compared to the initial frame



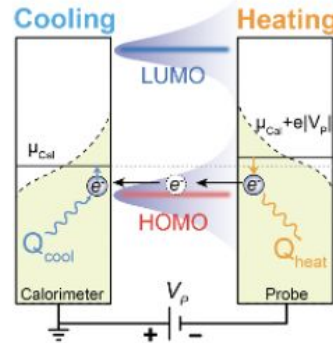
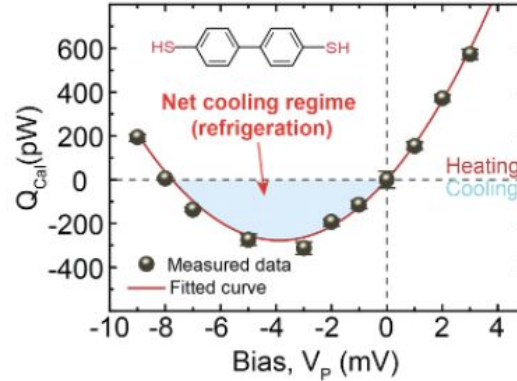
- Non-affine displacement:** to detect local structural rearrangement, we quantified the local non-affine displacement⁴ using D_{min}^2 :

$$D_{min}^2(t, \Delta t) = \frac{1}{n} \sum_{i=1}^n [\vec{r}_{ij}(t + \Delta t) - E\vec{r}_{ij}(t)]^2$$

Best fit affine transformation matrix

Probability density

- **Refrigeration at high density ($\sim 3 \text{ kW/cm}^2$) observed in Biphenyldithiol and explained in Landauer quantum transport picture**



Landauer theory:

$$Q(V_p) = \frac{2}{h} \int_{-\infty}^{\infty} (\mu_{\text{Cal}} - E) \tau(E, V_p) [f_{\text{Cal}} - f_p] dE$$

$$= GTSV_p + \frac{1}{2}GV_p^2 + O(V_p^3)$$

Q : Heating/cooling power V_p : Bias across molecules

μ : Chemical potential E : Energy level

f : Fermi-Dirac function G : Electrical conductance

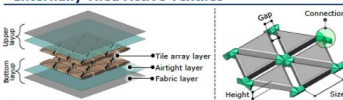
S : Seebeck coefficient T : Absolute temperature

- Net cooling (thermoelectric refrigeration) is accomplished when Peltier cooling is larger in magnitude than Joule heating effect.
- Cooling density sensitively depends on the electron transport characteristics.

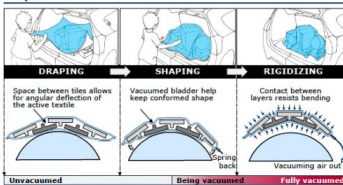
Goal

To develop an approach for understanding the user experience of an active textile with a tailored performance, which is enabled by a model-based performance balancing methods.

Internally Tiled Active Textiles



Operation

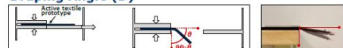


Fabrication

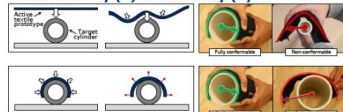


Test Setup and Procedure

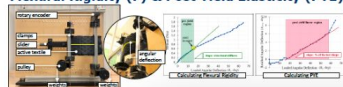
Draping Angle (D)



Conformability (C) & Settability (S)

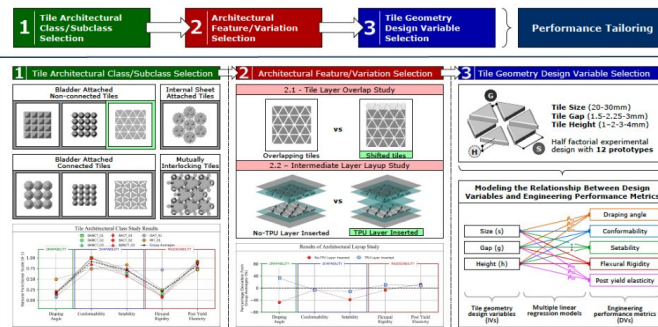


Flexural Rigidity (F) & Post Yield Elasticity (PYE)



Sequential Approach For Design Downselection

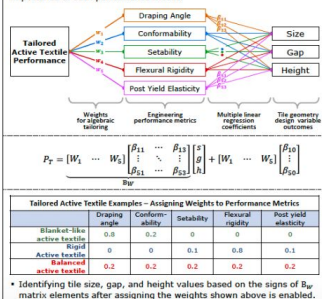
After defining all the engineering performance metrics, an in-depth understanding of the effects of tile architectural classes, the architectural features and metrics, and the tile geometry design variables on the active textile's overall performance are built, enabling performance tailoring technique.



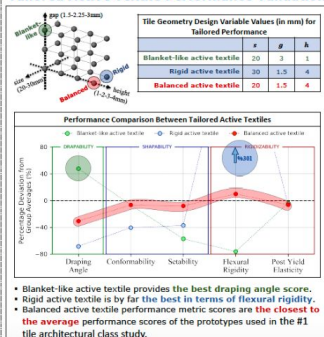
- The bladder attached non-connected tiles class, triangular tiles is downselected due to:
 - the lowest negative correlation among performance metric pairs
 - low profile cross section, effective packaging
- 24% increase in active textile performance metric scores on average by:
 - Shifting the tile array layers relative to each other
 - having an intermediate layer in-between
- 17% error on average for predicting two additional active textiles' performance
 - Increasing the sample size used to construct the model can improve the prediction error rate

Model-Based Performance Tailoring Approach

In design process, the goal is not just to predict the overall performance as an outcome, but to tailor the design and balance the relative importance of each performance metric.

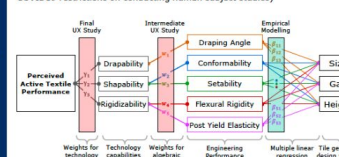


Tailored Active Textile Performance Validation



Understanding User Experience

To develop an understanding of the user experience, there will be two sets of user experience studies: Intermediate and final UX studies (pending due to COVID19 restrictions on conducting human subject studies)



Perceive Active Textile Performance

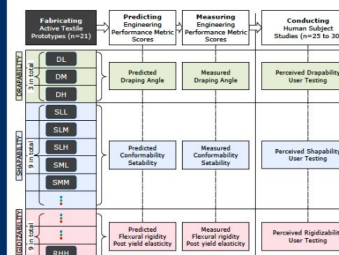
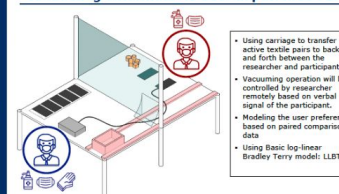
$$P_T = Drapability + r_1 \cdot Shapability + r_2 \cdot Rigidity$$

$$Drapability = w_1 \cdot draping_angle(x, g, h)$$

$$Shapability = w_2 \cdot conformability(x, g, h) + w_3 \cdot settability(x, g, h)$$

$$Rigidity = w_4 \cdot flexural_rigidity(x, g, h) + w_5 \cdot post_yield_elasticity(x, g, h)$$

Conducting Intermediate User Experience Test



Findings/Results present the facts

Result

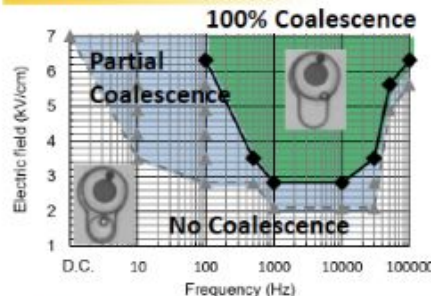


Figure 7: Observed coalescence response under different electric field strength and frequency. The green shaded area indicates all the droplets pairs are merged. Only a fraction of droplets is merged in the blue shaded area. No merging event occurred in white area. The electric field is calculated by the applied voltage V across two electrodes with a separation distance of 15mm ($E=V/0.15$, kV/cm).

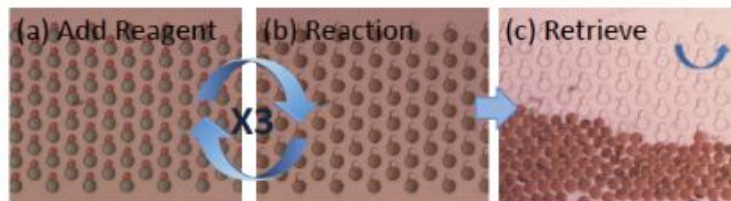


Figure 8: The capability of merging multiple droplets and retrieving them for down-stream analysis.

2-Bead Co-encapsulation

Poisson ($\lambda=0.1$)

Poisson ($\lambda=1$)

This work ($\lambda=0.05$)

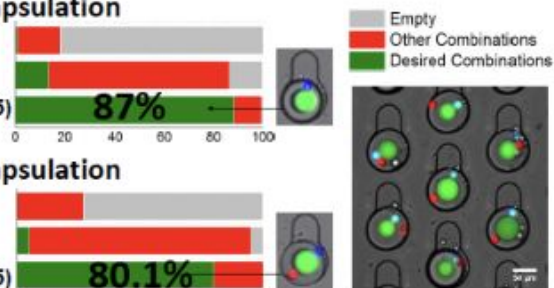


3-Bead Co-encapsulation

Poisson ($\lambda=0.1$)

Poisson ($\lambda=1$)

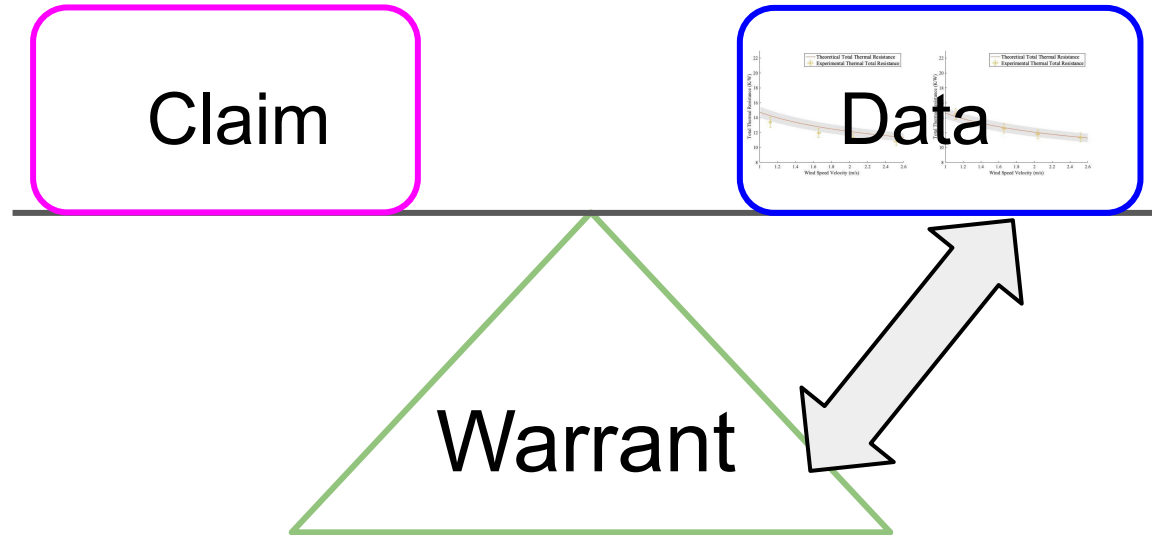
This work ($\lambda=0.05$)



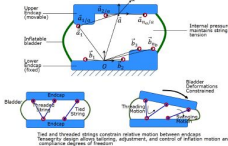
Percentage of Droplets (%)

Figure 9: The histogram shows the success rate of droplets that contain the correct numbers of blue, red, and green fluorescent beads. The data was collected from a total 1176 host droplets in a merging device with 3 repeats.

1045 medium carbon steel is harder than 1008 low carbon steel at the room temperature. The data from Table 1 show the average hardness of 1008 steel coupons is 85 HRB while that of 1045 steel is 99 HRB. Addition of carbon as an alloying element in ferrous alloys can create local nonuniformity in the lattice, which makes plastic deformation more difficult by impeding dislocation motion [1].



Tensegrity Concept



3D Kinematic Modeling

Small Deflection Kinematics

Local motions $\vec{v}$ and $\vec{\omega}$ where $\vec{d} = \begin{bmatrix} d_x \\ d_y \\ d_z \end{bmatrix}$ and $\vec{\omega} = \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$

Rate of change of string segment $(\vec{d}, \vec{\omega})$ length:

$$\frac{d}{dt} \left[\frac{1}{L} \left(\vec{d} \cdot \vec{d} \right) \right] = \frac{1}{L} \left(\vec{d} \cdot \frac{d\vec{d}}{dt} \right) = \frac{1}{L} \left(\vec{d} \cdot \vec{\omega} \right) \quad \text{where } \vec{d} \cdot \vec{\omega} = \vec{d} \cdot \vec{\omega}$$

Motion Constraints

- A single string connects pairs $(\vec{p}_i, \vec{p}_j)$ of points in opposite vertices
- Compresses sum of lengths to be constant

$$\sum_{i,j} \left[\frac{1}{L_{ij}} \left(\vec{p}_i - \vec{p}_j \right) \cdot \vec{v} \right] = 0$$

Each $\vec{p}_i$ generates a 1-N constraint vector $\vec{C}_{ij}$ the string restricts to perpendicular motion to $\vec{C}_{ij}$

Define constraint matrix $\mathbf{C}$ with rows $\vec{C}_{ij}$

$$\mathbf{C} = \begin{bmatrix} \vec{C}_{12} \\ \vdots \\ \vec{C}_{ij} \\ \vdots \\ \vec{C}_{nm} \end{bmatrix}$$

Instantaneous Centers

ICs are the intersection of the lines through the points of attachment of the strings

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CONTROLLED COMPLIANCE

Tailored Adjustable

Tensegrity designed to allow desired degrees of freedom and stiffness

Example Applications

- Inflatable Steering
- Deployable accessory controls
- Load-limited collapsible air dam
- Seating / head, foot, arm rest
- Deployable desk
- Fully rigid structures

Inflatable Steering Allows torsion, constrains other DOF

Load-Limited, Collapsible Inflatable Air Dam

Multi-Mode Deployable Button Controls

Moving Tensegrity Adjusts Steering Feel

Tensegrity guides inflation deployment

Bladder-Actuated

USEFUL OPERATION MODES

Low-Force: POSEABLE

Threading friction holds intermediate positions

Operating Principle

Example Applications

- Privacy screen
- Sun shade
- Cup holder
- Display interface

Manually Adjustable Privacy Screen

Clamps selectively lock threaded degrees of freedom

Operating principle

Deployable, Fold-Down Table

Actuated tensegrity strings change inflatable shape

Operating principle

Deployable, Movable Rear Spoiler

Operating principle

Operating principle

Operating principle

Operating principle

Operating principle

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Operating principle

Preliminary Experimental Case Study

Experimental Approach

Prototype Construction

Deflection Measurement

Shear x Shear y Axial z Translation DOF

Bend x Bend y Rotation DOF

Unconstrained Fully Constrained Parallel Constraint

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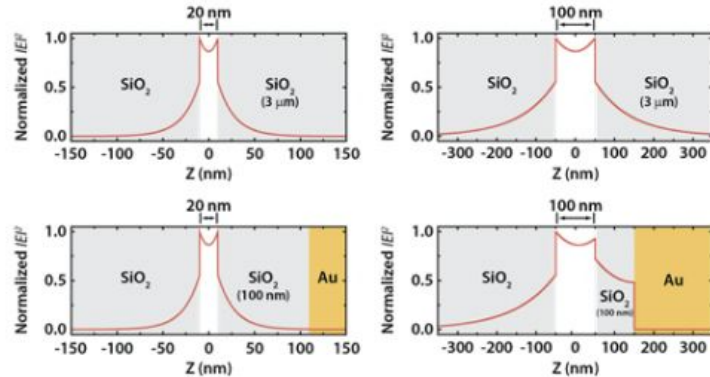
Unconstrained Fully Constrained Parallel Constraint

Unconstrained Fully Constrained Parallel Constraint

Unconstrained Fully Constrained Parallel Constraint

Your recommendations should consider your goals *and* findings

Penetration Depth Scales with Gap-Size



✓ For small gaps:

- Mode shapes are identical for different film thicknesses
- Penetration depth is the same order as the gap-size
- The modes decay rapidly into the film

✓ For larger gaps (greater than film thickness):

- Penetration depth becomes larger
- The modes decay slowly into the dielectric coating
- There is a sudden drop in intensity at Au-SiO₂ interface resulting in a diminished heat transfer at the interface

Note: your task letter is different; don't try to replicate

Linear and Nonlinear Models Drive Design Recommendations

Linear Model Predicts Adjustable Parameters With Greatest Sensitivity to System Behavior



Nonlinear Model Predicts Recommendations to Reduce Passenger Discomfort due to Vibrations

1:

Increase by 22%
= Lowers Peak

2: Wheel Inertia (J_d)

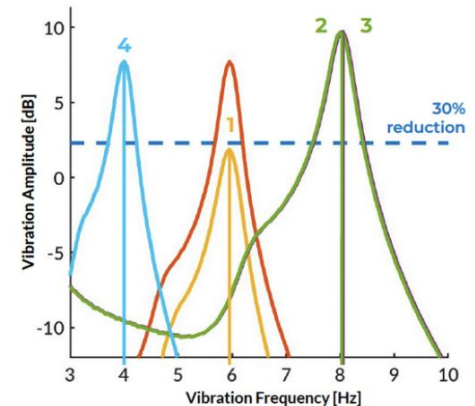
Decrease D
= Shifts Peak out of
Discomfort Region

3: Wheel Inerti
Reduce Inertia
= Shifts Peak o
Discomfort Region

4:

18%

Nonlinear Model Validates Recommendations to Reduce Passenger Discomfort due to Vibrations



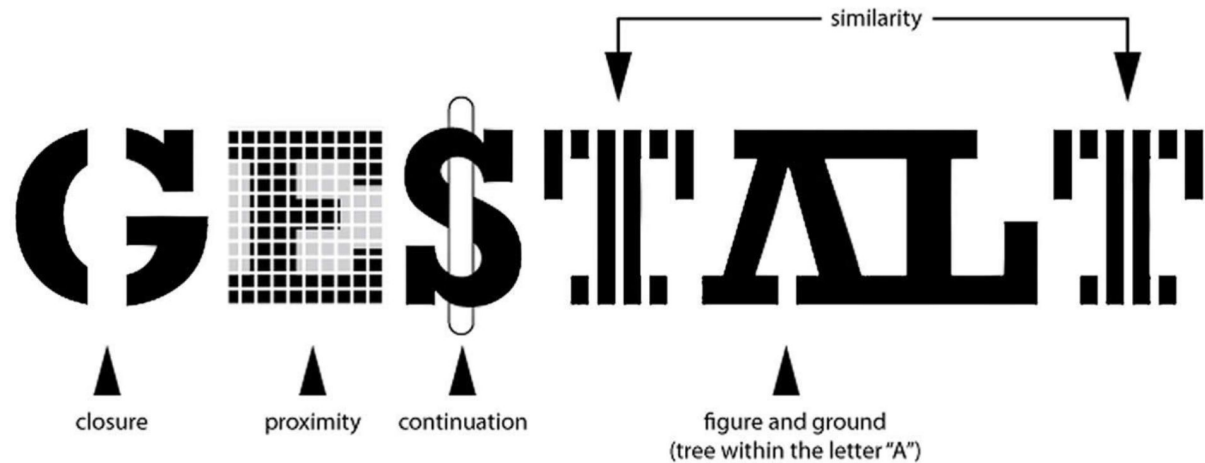
Let's examine design features

Closure

Color

Symmetry

White space



Understanding Mechanical Aging of a Granular Raft Under Small Amplitude Oscillatory Deformation

Will Hobson-Rhoades¹, Douglas Durian², Hongyi Xiao¹

¹Department of Mechanical Engineering, University of Michigan, Ann Arbor, ²Department of Physics and Astronomy, University of Pennsylvania

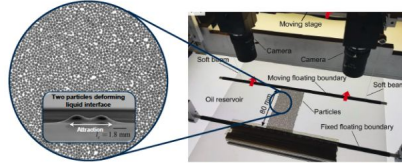
Motivation

Granular materials are well-known for their complex mechanical behaviors in response to applied loads. Periodically deforming granular matter below a strain threshold could lead to completely reversible particle trajectories¹. However, this is not always true for a system with a complex energy landscape², as the particle configuration could continuously evolve towards lower energy basins, which can be called aging. In this work, we aim to understand the long-term dynamics and structural evolution of a periodically deformed material with volumetric changes permitted. We perform such an experiment with particle-scale resolution using a two-dimensional particle raft, which is an important granular system that is widely seen in nature and industry.

Experimental setup and procedure

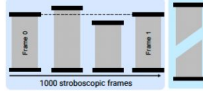
System setup:

- Monolayer of polydisperse Styrofoam particles floating at an air-oil interface. Long-ranged capillary attractions between particles exist³, with a characteristic interaction range of the capillary length $l_c = \sqrt{\gamma/(\rho g)}$, with γ being the surface tension, ρ the density of the liquid, and g the gravitational acceleration.
- Displacement applied with floating boundaries with a soft cantilever setup to measure the applied forces. Particle centers and radii were tracked⁴.



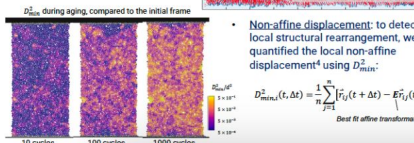
Quasi-static testing procedure:

- Mechanical aging:** small amplitude oscillatory deformation was applied for 1000 cycles with 0.3% strain amplitude
- Uniaxial deformation:** tension or compression is applied to the system until failure, with a strain rate on the order of 10^3 that minimize viscous drag⁵



Dynamics during mechanical aging

- Displacement field:** in stroboscopic images, particles have uneven convection-like current during the oscillatory deformation
- Particle trajectories** are irreversible in all oscillatory cycles, resulting in long-term non-affine displacement



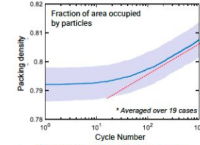
- Non-affine displacement:** to detect local structural rearrangement, we quantified the local non-affine displacement⁶ using D_{non}^2 :
$$D_{non}^2(\mathbf{r}, \Delta t) = \frac{1}{N} \sum_i \left| \mathbf{r}_i(t + \Delta t) - \mathbf{E} \mathbf{r}_i(t) \right|^2$$

Best fit affine transformation matrix

Structural change during mechanical aging

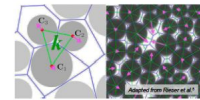
We examined the change of a few important structural descriptors during the mechanical aging process as a function of cycle number:

Density:



- The packing density increases continuously, signaling aging

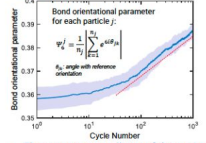
Heterogeneity:



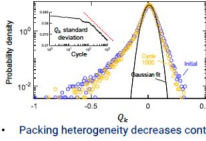
$$Q_k \equiv \langle \nabla \cdot \mathbf{c} \rangle (A_k / \bar{A})$$

- The logarithmic structural changes in the long term indicate increasing stability

Crystallinity:



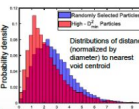
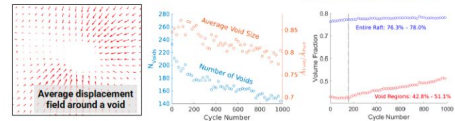
- The average crystallinity of the system increases with the cycle number



- Packing heterogeneity decreases continuously

Local dynamics driven by void collapsing

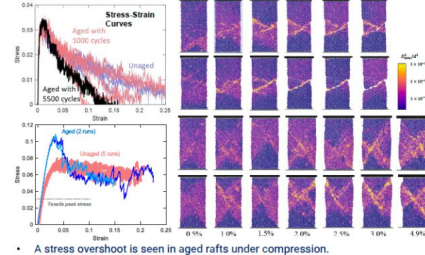
- Voids were identified via a morphological closing operation based on an image rendered using particle centers and radii. A threshold was chosen so that identified voids have larger areas than the smallest particles in the system.
- The displacement during aging evolves around void closure, which can be seen from the averaged displacement field centered around voids.



- The regions around the voids become roughly 1% more dense, compared to a 2% density increase of the entire raft
- Particles closer to the identified voids experience larger degree of rearrangement on average, which possibly contribute to densification, increasing crystallinity, and decreasing heterogeneity of the structure.

Aging-Induced Ductility Change

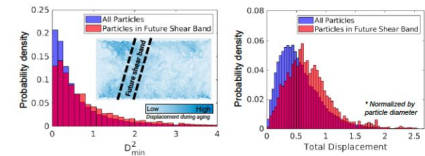
- Aged rafts tend to show a stronger degree of strain localization during subsequent uniaxial tests.



- A stress overshoot is seen in aged rafts under compression.

Precursors for the shear band

- Shear bands tend to form along regions of the raft that experience high displacement during aging. On average, particles that will be present in the shear bands travelled further as the system aged



- We are currently looking for how corresponding structural signals, i.e., weak spots, evolve over the mechanical aging process.

Summary

- A granular raft ages under periodic uniaxial mechanical deformation with continuously evolving dynamics and structure
- The aging process, reflected by density increase and heterogeneity decrease, is driven by a large-scale displacement field driven by void closure events
- The aged rafts demonstrate higher degree of strain localization during subsequent tensile and compression tests

References

- [1] Zhao, Yiqiu, et al. "Microscopic reversibility and emergent elasticity in ultrastable granular systems." *Frontiers in Physics* 10 (2022): 1049683.
- [2] Yuan, Ye, et al. "From creep to flow: Granular materials under cyclic shear." *Nature Communications* 15.1 (2024): 3856.
- [3] Xiao, Hongyi, Robert J. Ivancic, and Douglas J. Durian. "Strain localization and failure of disordered particle rafts with tunable ductility during tensile deformation." *Soft Matter* 16.35 (2020): 8226-8236.
- [4] Falk, Michael L., and James S. Langer. "Dynamics of viscoplastic deformation in amorphous solids." *Physical Review E* 57.6 (1998): 7152.
- [5] Rieser, Jennifer M., et al. "Divergence of Voronoi cell anisotropy vector: a threshold-free characterization of local structure in amorphous materials." *Physical Review Letters* 116.8 (2016): 088001.

No Strings Attached: A Sensor-Based Pedal Steel Guitar

ECCS 452: Digital Signal Processing Design Lab
Student Name, Student Name, Student Name, and Student Name

INTRODUCTION

Our project is an electronic pedal steel guitar, a low-cost, sensor-based alternative to a traditional pedal steel guitar. To play a traditional pedal steel, a performer plucks the guitar strings, moves a metal slide up and down the strings to control their pitches, and "bends" the pitches of certain strings with foot pedals. Using sensors to electronically emulate these functionalities, our instrument is novel, easy to play, and is significantly less expensive than a traditional pedal steel guitar. Our demonstration showcases our instrument's capabilities with a fun musical experience.



Figure 1. Major hardware components of the electronic pedal steel guitar.

SYSTEM REQUIREMENTS

- Inexpensive ("\$300) in comparison to the price of traditional pedal steel guitars (\$1000-\$4000 and up)
- Work in real-time without noticeable delay
- Sensitive to subtle changes in sensor input
- Sound similar to a traditional pedal steel guitar
- Robust to potential rough handling

DESIGN CHALLENGES

To keep prices low, we used a combination of low-cost sensors and wires to collect user input as well as 3D-printed custom parts.

To speed up our algorithms, we used several optimization techniques, including pre-calculating and storing parameters and multithreading code.

OVERALL SYSTEM ARCHITECTURE



Figure 2. Overall system architecture: An array of sensors feed data to a pair of Raspberry Pi working in synchrony to synthesize a guitar-like sound in response to user inputs.

HARDWARE ARCHITECTURE



Figure 3. Test setup. Five major groups of sensors are used in our system: ribbon sensors for the fretboard (left), vibration sensors for pluck detection (bottom right), a pressure sensor for muting (middle right), a circular position sensor for volume control (top right), and two potentiometers for bending pitches (not shown). The sensors interface to the first Raspberry Pi (middle) through a pair of analog-to-digital converters (ADCs). The first Pi communicates with the second Pi (top left) via an Ethernet connection, and the second Pi outputs the generated audio.

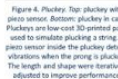


Figure 4. Pluck test setup. Five major groups of sensors are used in our system: ribbon sensors for the fretboard (left), vibration sensors for pluck detection (bottom right), a pressure sensor for muting (middle right), a circular position sensor for volume control (top right), and two potentiometers for bending pitches (not shown). The sensors interface to the first Raspberry Pi (middle) through a pair of analog-to-digital converters (ADCs). The first Pi communicates with the second Pi (top left) via an Ethernet connection, and the second Pi outputs the generated audio.

SOFTWARE ARCHITECTURE

Fretboard: sensor readings are calibrated to fit non-linearities then converted to frequencies to emulate a guitar fretboard (Figure 5).

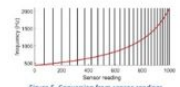


Figure 5. Conversion from sensor readings to frequencies on our 27 fret fretboard.

Pluck detection: a modified smoothed 2-score algorithm [2] detects spikes in the voltage output corresponding to plucks, ignoring spikes that are too small or too close together.

Volume control: audio gain is scaled exponentially by a smoothed derivative of the circular ribbon sensor output. To cancel the effects of noise, derivatives that are too high or too low are ignored.

Sound generation: The Karplus-Strong algorithm (Figure 6) mimics the sound of plucked strings, using low-pass filters to attenuate the high frequency sound components in a feedback loop [1].

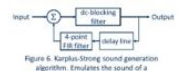


Figure 6. Karplus-Strong sound generation algorithm. Emulates the sound of a plucked instrument.

RESULTS

Our system synthesizes the clean, sustained sounds of a steel guitar in real-time, producing smooth variations in pitch that match the fretboard position with the slide. The system detects plucks with virtually no false positives or false negatives (Figure 7).

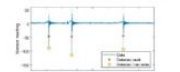


Figure 7. Results of pluck detection algorithm.

Our system requires 10 µs to synthesize one audio sample, twice as fast as required for our sample rate of 48 kHz. The volume control algorithm varying gain that plateaus at a set minimum and maximum gain (Figure 8).

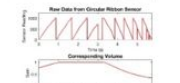


Figure 8. Results of volume control algorithm.

ACKNOWLEDGEMENTS

Our team gratefully acknowledges the help and funding that we received from the Harris Corporation, the ECC Undergraduate Program, the U-M 3D Printing Studio Staff, Professor Gregory Wakefield, Professor Kurt Metzger, and Siddharth Venkatesan.

CITATIONS

- [1] D. J. Thomson, "The Karplus-Strong algorithm for digital string synthesis," in *Proceedings of the 1980 IEEE Conference on Acoustics, Speech, and Signal Processing*, vol. 1, pp. 319-322, 1980.
- [2] J. R. D. Jones, "A modified smoothed 2-score algorithm for pluck detection," in *Proceedings of the 2010 IEEE Conference on Acoustics, Speech, and Signal Processing*, pp. 4454-4458, 2010.

No Strings Attached: A Sensor-Based Pedal Steel Guitar

ECCS 452: Digital Signal Processing Design Lab
Student Name, Student Name, Student Name, and Student Name

INTRODUCTION

Our project is an electronic pedal steel guitar, a low-cost, sensor-based alternative to a traditional pedal steel guitar. To play a traditional pedal steel, a performer plucks the guitar strings, moves a metal slide up and down the strings to control their pitches, and "bends" the pitches of certain strings with foot pedals. Using sensors to electronically emulate these functionalities, our instrument is novel, easy to play, and is significantly less expensive than a traditional pedal steel guitar. Our demonstration showcases our instrument's capabilities with a fun musical experience.



Figure 1. Major hardware components of the electronic pedal steel guitar.

SYSTEM REQUIREMENTS

- Inexpensive ("\$300) in comparison to the price of traditional pedal steel guitars (\$1000-\$4000 and up)
- Work in real-time without noticeable delay
- Sensitive to subtle changes in sensor input
- Sound similar to a traditional pedal steel guitar
- Robust to potential rough handling

DESIGN CHALLENGES

To keep prices low, we used a combination of low-cost sensors and wires to collect user input as well as 3D-printed custom parts.

To speed up our algorithms, we used several optimization techniques, including pre-calculating and storing parameters and multithreading code.

OVERALL SYSTEM ARCHITECTURE

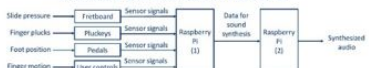


Figure 2. Overall system architecture: An array of sensors feed data to a pair of Raspberry Pi working in synchrony to synthesize a guitar-like sound in response to user inputs.

HARDWARE ARCHITECTURE



Figure 3. Test setup. Five major groups of sensors are used in our system: ribbon sensors for the fretboard (left), vibration sensors for pluck detection (bottom right), a pressure sensor for muting (middle right), a circular position sensor for volume control (top right), and two potentiometers for bending pitches (not shown). The sensors interface to the first Raspberry Pi (middle) through a pair of analog-to-digital converters (ADCs). The first Pi communicates with the second Pi (top left) via an Ethernet connection, and the second Pi outputs the generated audio.

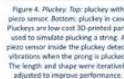


Figure 4. Pluck test setup. Five major groups of sensors are used in our system: ribbon sensors for the fretboard (left), vibration sensors for pluck detection (bottom right), a pressure sensor for muting (middle right), a circular position sensor for volume control (top right), and two potentiometers for bending pitches (not shown). The sensors interface to the first Raspberry Pi (middle) through a pair of analog-to-digital converters (ADCs). The first Pi communicates with the second Pi (top left) via an Ethernet connection, and the second Pi outputs the generated audio.

SOFTWARE ARCHITECTURE

Fretboard: sensor readings are calibrated to fit non-linearities then converted to frequencies to emulate a guitar fretboard (Figure 5).

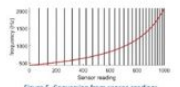


Figure 5. Conversion from sensor readings to frequencies on our 27 fret fretboard.

Pluck detection: a modified smoothed 2-score algorithm [2] detects spikes in the voltage output corresponding to plucks, ignoring spikes that are too small or too close together.

Volume control: audio gain is scaled exponentially by a smoothed derivative of the circular ribbon sensor output. To cancel the effects of noise, derivatives that are too high or too low are ignored.

Sound generation: The Karplus-Strong algorithm (Figure 6) mimics the sound of plucked strings, using low-pass filters to attenuate the high frequency sound components in a feedback loop [1].



Figure 6. Karplus-Strong sound generation algorithm. Emulates the sound of a plucked instrument.

RESULTS

Our system synthesizes the clean, sustained sounds of a steel guitar in real-time, producing smooth variations in pitch that match the fretboard position with the slide. The system detects plucks with virtually no false positives or false negatives (Figure 7).

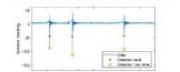


Figure 7. Results of pluck detection algorithm.

Our system requires 10 µs to synthesize one audio sample, twice as fast as required for our sample rate of 48 kHz. The volume control algorithm varying gain that plateaus at a set minimum and maximum gain (Figure 8).

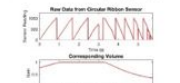


Figure 8. Results of volume control algorithm.

ACKNOWLEDGEMENTS

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CITATIONS

- [1] D. J. Thomson, "The Karplus-Strong algorithm for digital string synthesis," in *Proceedings of the 1980 IEEE Conference on Acoustics, Speech, and Signal Processing*, vol. 1, pp. 319-322, 1980.
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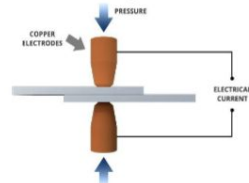


Welding Design Changes for EV Battery Packs using Parameters from Subassembly

EV POWER PROVIDERS, INC.

Background

EV Power Providers, Inc. has requested a next-generation battery pack design change via spot welding; a process in which applies pressure and current to objects, generating heat to secure them together.



[https://www.twi-global.com/technical-knowledge/faqs/what-is-spot-welding#:~:text=Spot%20welding%20\(also%20known%20as,current%20to%20the%20weld%20area.](https://www.twi-global.com/technical-knowledge/faqs/what-is-spot-welding#:~:text=Spot%20welding%20(also%20known%20as,current%20to%20the%20weld%20area.)

Objectives

- Determine optimal weld conditions and resulting force values for differing constraints
- Determine relationships between the data to improve nondestructive evaluation methods

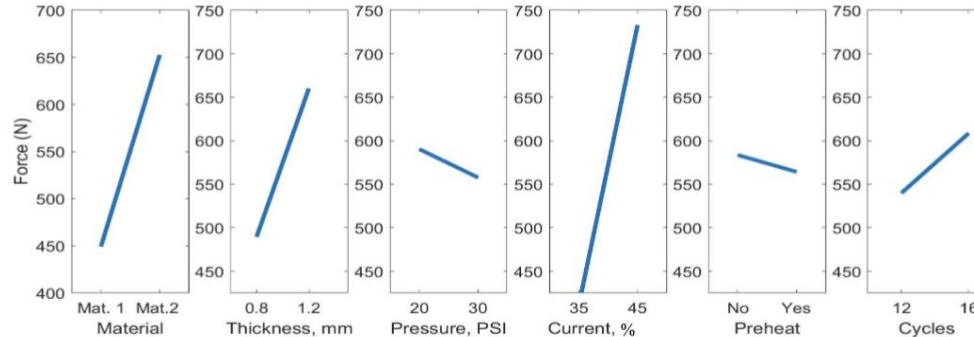
Methods

- Solved for 6 optimal conditions of a weld through experimental design
- Determined sub-optimal conditions for a weld of 1.2 mm galvanized steel with no preheat and 30 psi of applied pressure
- Deduced the viability of resistance as a function of non-destructive evaluation of weld quality
- Found the maximum tensile force and explained how tensile test data could be used to find the failure and yield strengths of the weld

References

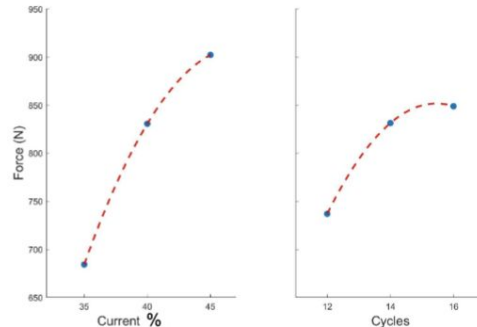
Smith, S. (n.d.). What is spot Welding? (a complete welding Process Guide). Retrieved March 15, 2021, from [https://www.twi-global.com/technical-knowledge/faqs/what-is-spot-welding#:~:text=Spot%20welding%20\(also%20known%20as,current%20to%20the%20weld%20area.](https://www.twi-global.com/technical-knowledge/faqs/what-is-spot-welding#:~:text=Spot%20welding%20(also%20known%20as,current%20to%20the%20weld%20area.)

Unconstrained Main Effects Analysis



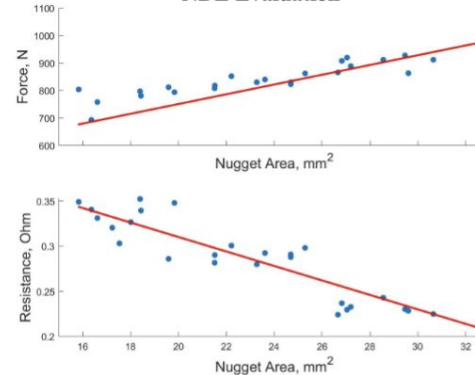
- Main effects plot displays 6 conditions and their effect on weld strength. Mat. 1 is galvanized and Mat. 2 is galvanized.

Constrained Main Effects Analysis



- Highest tensile forces generated in the constrained case during applied conditions: 45 % and 16 cycles applied to the weld

NDE Evaluation



- Inverse relationship between nugget area and resistance ($R^2 = 0.7284$). Direct relationship between nugget area and force ($R^2 = -0.9$). No correlation was found between force and resistance; coefficient of 0.16.

4. Maximum Tensile Force

Unconstrained	584.52 ± 56.50 N
Constrained	805.63 ± 24.64 N

- Constrained samples test the effects of current and number of cycles on a non-preheated galvanized steel at 1.2mm thickness under 30psi.
- Use tensile test data to find failure strength by finding fracture toughness via a requested load vs displacement curve
 - Integrate to find energy = fracture toughness
- Request strain data from tensile test to determine the yield force
 - Assume 2 % increase from initial to yield elongation
 - Time marker to find the instantaneous force (yield force)

Conclusion / Recommendations

- Six optimal conditions for weld strength in the constrained (805.63 N) and unconstrained (584.52 N) cases
- Nugget area allows a relationship between resistance and force to be established.
- Additional testing is needed under constrained conditions with 20 PSI applied pressure.
- Optimal weld conditions based on analysis are galvanized, 1.2 mm, 20 psi, 45 Amps, preheated, 16 cycles

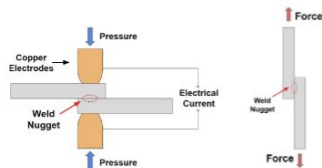
Energy is a Valuable Parameter to Determine Spot Weld Quality

Background and Objectives

Resistance spot welding is a process in which two or more metal sheets are pressed together by electrodes. A current is applied through the plates to generate heat and melt the sheets together. To determine the weld conditions that result in the highest quality welds for EV Power Provider's tab-to-terminal connections, the R&D department has been tasked with performing a Design of Experiments (DOE) analysis of several weld factors. For these experiments, the welds were generated between two sheets of steel using the T3 Snow resistance welder, and strength tested with the Instron 4466 load frame. This presentation describes the analysis process, presents the data evaluation, and discusses the highest quality welds conditions.

Our tasks are as follows:

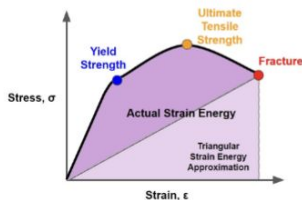
- Determine the optimal weld conditions
- Determine the suboptimal weld conditions of 1.2 mm galvanized steel with no preheat and pressure equal to 30 psi
- Analyze the usefulness of measuring resistance as a method of predicting weld quality through non-destructive evaluation (NDE)
- Discuss how tensile test data can be used to find the yield and failure strengths
- Explain future testing recommendations



Diagrams showing the setup of a resistance spot welding rig, and a welded sample under tension.

Yield and Failure Strength Of Welds

The failure strength of each weld can be determined with the maximum force each spot-welded sample could withstand, and the measured nugget area. Since strain data was not collected during this round of testing, it was not possible to determine the yield strength of each sample. If strain data was collected, the yield strength could be found by determining the stress after which the sample takes on a permanent strain of 0.002 when unloaded.



Example stress-strain curve, highlighting key points of mechanical behavior.

Optimal Weld Condition

Varying combinations of weld parameters and subjecting the different samples to tensile tests allowed for the determination of optimal weld condition and the associated max tensile force.

A linear regression between each parameter and maximum tensile force was created, and for an individual parameter, the parameter state that provided the maximum tensile force was considered ideal.

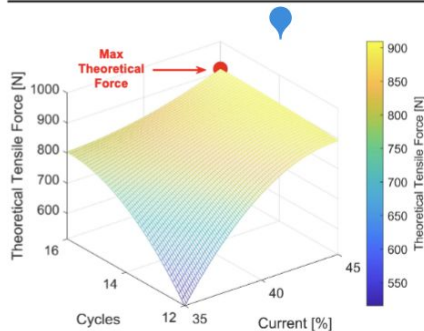
Sub-Optimal Weld Condition 1.2 mm Galvanized Steel

Under suboptimal conditions, the same procedure of varying parameters and tensile testing was followed. Fewer parameters were varied with an additional level of variation to the state of each individual parameter. This finer mesh lead to different results of max tensile force, compared to the optimal conditions.

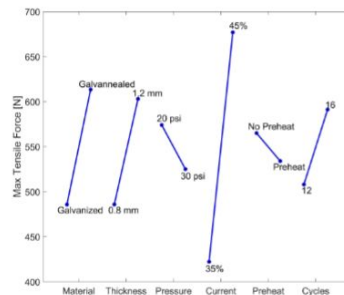
In order to isolate the influence of current and cycles on the main factors of weld quality (tensile force and nugget diameter/area), bi-quadratic regressions on the data were performed. The results of this theoretical regression model were compared to the experimental results of maximum tensile force and maximum nugget area.

Table summarizing results of optimal and suboptimal weld conditions.

	Optimal Condition Tensile Force [N]	Suboptimal Condition Tensile Force [N]	Suboptimal Condition Nugget Area [mm ²]
Experimental Data	893 N ± 39 N	914.7 N ± 7.5 N	30.6 mm ² ± 3.1 mm ²
Theoretical Model	891 N ± 190 N	910 N ± 100 N	30.4 mm ² ± 4.7 mm ²



Graph displaying a theoretical bi-quadratic regression model of tensile force, current, and cycles.

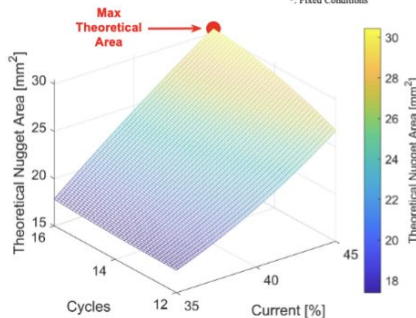


DOE analysis used to determine optimal weld conditions.

Table showing the best conditions for max tensile force experimentally.

Parameter	Optimal Condition	Suboptimal Condition
Material	Galvanized	Galvanized*
Thickness	1.2 mm	1.2 mm*
Pressure	20 psi	30 psi*
Current	45%	45%
Preheat	No	No*
Cycles	16	14

*: Fixed Conditions



Graph displaying a theoretical bi-quadratic regression model of nugget area, current, and cycles.

Non-Destructive Evaluation

Predictability of weld quality can be achieved by analyzing weld parameters non-destructively. Weld resistance, calculated from the weld current and voltage, was investigated as a possible non-destructive indicator of weld quality with mixed results. Resistance was found to have a moderate correlation ($r = -0.59$) with maximum tensile force and a strong correlation ($r = -0.90$) with nugget area [1]. Because max tensile force is a stronger indicator of weld quality, the correlation of different parameters to tensile force and nugget diameter/area were evaluated. The parameters of current, power, and input energy all showed stronger correlations with tensile force and nugget area and are thus more accurate factors to use in an NDE of weld quality.

Table displaying further analysis of current, power, and energy, comparing correlation coefficient results (R) with resistance.

NDE Parameter	R Value (Tensile Force)	R Value (Nugget Area)
Resistance	-0.59	-0.90
Current	0.75	0.93
Power	0.82	0.92
Energy	0.84	0.90

Recommendations

From our analysis of the DOE data, we have a number of recommendations for future testing that would expand upon the results of the current data.

- Test a greater range of cycles
- Test larger input currents
- Conduct in-depth investigation into new possible NDE indicators
 - Input current
 - Input power
 - Input energy
- Collect strain data from tensile tests
 - Determine yield strength
 - Determine strain energy
 - Determine correlation between input energy and yield strength

References

1. Moore, D. S., Notz, W. I., & Flinger, M. A. (2013). The basic practice of statistics (6th ed.). New York, NY: W. H. Freeman and Company.
2. Kou, Sindo, (2020). Welding Metallurgy. Hoboken, NJ: John Wiley & Sons Inc.

Poster Checklist

1. Visual Message
2. Title makes a claim and/or tells us the result
3. Text = 30/40% (text/labels are large enough, use margins & white space)
4. Supporting visuals = 60/70%
5. Limited branding
6. Descriptive headings, logical flow and order (guide the eye)
7. Avoid non-data ink
8. Visuals: resolution & clarity
9. Design so good you don't know it's there

Submit to 3 places:
GSI, and Canvas
–TC and Tech
assignments

Submission Instructions

Due at the start of your lab session's [presentation schedule](#)

- Email your poster as a PDF to your GSI
- Submit your poster here as a PDF (do not submit handout)
- Submit your poster and handout to the technical assignment
- Print 4 8.5x11" copies of your poster (color preferred, but not required)
- Print 4 copies of your handout (color preferred, but not required; stapled *strongly* preferred)

Note that the Canvas due date is simply a placeholder to remind you to submit your poster here (in addition to submitting the poster and handout to the technical submission), rather than the actual due date.

Purpose

You will respond to a memo (or task letter) from your colleagues at a fictional company describing the organizational problem they are facing and how you can help them solve it. Most of the time in your career, when delivering a presentation, you'll have the opportunity to use slides or share your screen or use whatever other visual aid you want to. This assignment, however, presents a unique challenge. In this case, you have very limited space, enough for a few visuals and almost no text. Thus, you will need to analyze your audience's needs and prioritize certain data over others. You will also need to consider how to balance delivering content visually and orally while respecting your audience's cognitive load. You will need to anticipate tough questions from your audience.

Our hope is that, by challenging yourself to put forward an effective argument under such a strict space constraint, you will develop skills that will serve you well even when you have the option to spread out your argument over many slides.

This assignment addresses all of the ME495 Tech Comm learning objectives:

- Conduct audience analysis to determine effective communication strategies for diverse audiences.
- Construct persuasive arguments by organizing claims and supporting them with relevant and convincing evidence from experimental data and literature.
- Deliver clear, persuasive oral presentations of technical content.
- Design documents for usability and accessibility, and integrate visual aids to enhance the clarity and effectiveness of both written and oral communication.
- Engage deliberately and collaboratively in iterative writing and design processes that includes research, drafting, peer review, revision, and reflection.

TC Poster Rubric (2)

Criteria	Ratings				Points
Audience Awareness and Purpose view longer description	Exceeds 40 to >36 pts	Meets 36 to >32 pts	Approaching 32 to >24 pts	Does not meet 24 to >0 pts	-- /40 pts

Comment

Leave a comment

Clear

Visuals and Poster Conventions view longer description	Exceeds 40 to >36 pts	Meets 36 to >32 pts	Approaching 32 to >24 pts	Does not meet 24 to >0 pts	-- /40 pts
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Comment

Leave a comment

Clear

Delivery view longer description	Exceeds 20 to >18 pts	Meets 18 to >16 pts	Approaching 16 to >12 pts	No Marks 12 to >0 pts	-- /20 pts
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Comment

Leave a comment

Clear

Time limit view longer description	Full Marks 0 pts	No Marks 0 pts	-- /0 pts
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Comment

Resources are available!



Mike Morrison [Video](#)

[TC Office Hours](#)

<https://www.craftofscientificposters.com/>
(poster templates, examples, advice)

Anything by Nancy Duarte (not just posters, but visuals and any presentation--she contributes to a lot of blogs, writes short articles, easily digestible and USEFUL tips you can use today)

<https://www.duarte.com/>

Jean-Luc Doumont <https://www.principiae.be/book/X0100.php>

Edward Tufte (perhaps best known for his criticism of PowerPoint)

<https://www.edwardtufte.com/tufte/>

Watch Prof. Montgomery's lecture on effective data visualization



Search or ask a question

1. He's funny
2. The video is high quality
3. It's geared toward your needs
4. It will teach you how to think about and plan visuals

Height of female popstars:

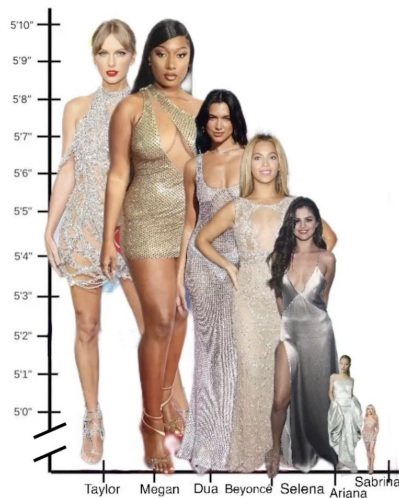


Image from Reddit

Technical Communication Lecture: Data Visualization

Unlisted



Joseph Montgomery
4 subscribers

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Share

93 views Jan 26, 2026

0:38 Relevant and Purposeful

9:49 Accurate and Ethical

23:25 Readable and Accessible

27:44 Minimal and Appealing

The Handout

Think of the handout as an appendix to the poster

Keep it highly visual but you can add paragraph-form text

Only your GSI's will grade it; consider your audience!

Include technical details they want to know

You can refer to the handout when you present your poster (bring printed copies!)

3 pages is the strict limit; you can do it!

Poster

Title that describes the result

Intro/background

Experiment setup

Design parameters

Model reliability

Show your work: calculating/testing/analysis

Multiple graphs/tables

Conclusions & Recommendations

References

Handout

Thumbnail of the poster

Multiple graphs/tables

Use headers/subheaders

Data/info that won't fit on the poster

All important info from the poster

References

TC will not grade the handout; meant for a tech audience

Please answer this survey on your way out:

**Poster
Workshop &
whole teaching
team Q&A: 9/18**



Poster Delivery

Posters require that you prepare pitches

Quick conference pitch—to the random audience member who may wish to move on

Tech-heavy pitch—to content-experts that are familiar with the jargon, acronyms, etc.

Non-tech pitch—to an educated audience with NO tech background

Practice Your Delivery...Yes, Actually Do This

Replicate the presentation scenario as much as possible
(in the lab)

Practice saying it out loud...a lot

Plan Your Introduction Carefully

- Practice your opening line WORD-FOR-WORD
- Introduce yourself
- Tell us why we are here
- Gesture toward the recommendation

Be Aware of Your Presentation Voice

- Speak slowly and intentionally
 - You speak faster when nervous
 - You know the material—we do not
- Pause
 - Give us time to process your material
 - Break at intentional moments
- Use emphasis for main points: *sound excited!*
 - Speak louder or pause
 - Use gestures

Use Transitions to Your Advantage

Use transitions effectively, “And now, Hector is going to discuss...”

“Thanks, Farah, I’m going to be speaking about...”

Use effective redundancy through transitions

- Helps structure your talk (serves as an agenda)
- Helps argue your point (tells us the argument in the transition)

“Now, Hector is going to explain why our proposed solution is the most effective...”

A generosity mindset can help you succeed

We physiologically register an audience as a threatening predator and mount a comparable response

The key to calming the amygdala and disarming our organic panic button is to turn the focus away from ourselves — away from whether we will mess up or whether the audience will like us — and toward helping the audience

Showing kindness and generosity to others has been shown to activate the vagus nerve, which has the power to calm the fight-or-flight response.

Practice generosity in planning and delivery

Think about your audience first, not the topic

What do they need? What can I teach them?

Focus on 1-3 people at a time

Make eye contact, focus, talk directly to *people*

Common Pitfalls

- There is no oral introduction: why are we all in this room?
- The solution is not indicated until the very end of the presentation
- There is a lack of flow or connection between parts of the presentation (too many assumptions)
- There is not a clear solution/conclusion presented

Content and Speech

Audience targeted?

Purpose achieved?

Needed information conveyed?

Terms defined; background given?

Emphasis appropriate?

Structure

Context?

Background and credibility?

Memorable mapping?

Depth satisfies?

Assertions emphasized?

Assertions supported?

Main assertions summarized?

Closure achieved?

Repetition used effectively?

Placement used effectively?

Visuals

Each section conveys a message?

Each section serves the audience?

Visual evidence supports?

Level of detail appropriate?

Visual evidence explained?

Extraneous details excluded?

Type easy to read?

White space used effectively?

Delivery

Speaker shows passion?

Speaker exudes confidence?

Voice clear and engaging?

Speed is appropriate?

Filler phrased ("uh") avoided?

Effective eye contact made?

Movements contribute?

Questions handled convincingly?

Time is appropriate?

Privilege clarity and organization

Focus on your core message

- Reduce text

- Keep it visual

- Use obvious sequence (place items where we expect to see them)

Maximize the signal-to-noise ratio

- Eliminate the noise that is in your control

- Use effective redundancy

Design Element	Poster Detail	Best Practice Design Elements
Text Content	Title	Concise Describes Outcome
	Author Info	Included
	Introduction	Brief Clear; often broken into two paragraphs Includes essential background, research aims & hypothesis References used
	Methods	Clear and concise Flowchart used
	Results	Concise with bullet points Graphical summary of main findings References
	Figure Legend	Included next to graph or not needed
	Table Legend	Brief
	Conclusion	Brief, restates major finding Describes limitations & future directions
	References	Essential only
	Acknowledgments	Brief
Graphic Content	Headings	Describes content in section All caps, numbered
	Diagrams	Used to graphically present methods Used to summarize major finding
	Table	Methods-clear
	Graphs	Large Chart junk removed No key Axis labeled Title describes what graph depicts No legend No background color
Aesthetics	Images/Logos	Logo small and at bottom Images large and high contrast
	Font Type	Sans serif for headings, serif for text Left justified
	Font Size	Minimum 24-point, font 'hierarchy' consistent
	Color Scheme	Cool tones that relate to the content
	Poster Background	White, all elements clear and stand out
	Layout (flow)	All elements aligned Ample white space Flow is clear from headings Results is largest section
	Overall Assessment	Mainly visual with clear graphics and good flow; little text used, high signal to noise ration

Sources

Alley, Michael. The Craft of Scientific Presentations : Critical Steps to Succeed and Critical Errors to Avoid. Springer New York, doi:10.1007/978-1-4419-8279-7.

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What the Customer Described.



What got budgeted.



What the Engineer Designed.



How Manufacturing Installed it.



What Marketing Advertised.



What got documented.



What the Customer finally received.



What the customer was billed for.



What the customer actually wanted.